

Bill of Materials

BOM

Component	Material	g/cell	kg/kWh	Formula / source
Positive electrode active material	NMC811 (90% of coating)	10.31	0.001	coating mass = $L \cdot A \cdot (1-\epsilon) \cdot \rho$ = cell/r1_v4_energy.json layer_kg_m2 $\times$ area; 90/5/5 active:CB:binder literature default (no binder/CB params)
Positive conductive additive	carbon black (5%)	0.57	0.000	literature default split, annotated
Positive binder	PVDF (5%)	0.57	0.000	literature default split, annotated
Negative electrode active material	graphite (95% of coating)	7.02	0.001	coating mass = $L \cdot A \cdot (1-\epsilon) \cdot \rho$; 95/2/3 active:CB:binder literature default
Negative conductive additive	carbon black (2%)	0.15	0.000	literature default split, annotated
Negative binder	PVDF/SBR (3%)	0.22	0.000	literature default split, annotated
Separator	polyolefin	0.22	0.000	cell/r1_v4_energy.json layer_kg_m2 separator $\times$ area
Electrolyte	EC/EMC + LiPF6 (high-transport)	4.72	0.000	pore volume $\times$ 1.2 g/cm ³ (literature electrolyte density, annotated); fill factor 1.0
Positive current collector	Al foil	2.22	0.000	cell/r1_v4_energy.json layer_kg_m2 positive_cc $\times$ area
Negative current collector	Cu foil	5.52	0.000	cell/r1_v4_energy.json layer_kg_m2 negative_cc $\times$ area
Enclosure	pouch/can	Not modeled	Not modeled	honest note: beyond parameter set
Tabs/terminals	–	Not modeled	Not modeled	honest note: beyond parameter set
TOTAL (incl. electrolyte)		31.5	0.002	electrolyte included BOM caliber; contract caliber mass = 26.8 g (excl. electrolyte/casing)
Cell energy		12.68		cell/r1_v4_energy.json:energy_wh